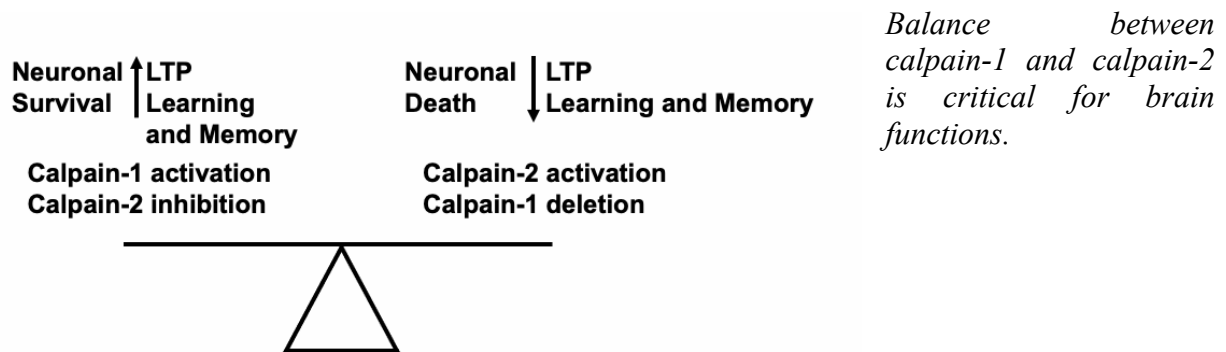


## Rationale for Selective Calpain-2 Inhibitors for the Treatment of Neurodegeneration

Our work over the last 45 years has clearly demonstrated that the balance between calpain-1 and calpain-2 activity in the brain is critical for both learning and memory and neuroprotection/neurodegeneration (see figure).



Importantly, calpain-1 activation or calpain-2 inhibition **enhances** learning and memory and neuronal survival, while calpain-2 activation or calpain-1 deletion/inhibition **limits** learning and memory and enhances neuronal death. Therefore, a selective calpain-2 inhibitor produces enhanced learning and promotes neuronal survival following a variety of injuries.

We have generated a series of novel calpain inhibitors and identified several with better selectivity for calpain-2 than for calpain-1. In particular, we believe selective calpain-2 inhibitors NA-184 and NA-112 are ready to be tested in humans for the treatment of several potential clinical indications.

The USPTO has recently validated our claims that these novel inhibitors prevent neurodegeneration (PCT/US2021/018042: CALPAIN-2 INHIBITOR COMPOUNDS AND METHODS OF TREATMENT; Notice of allowance dated 4/16/2025).

The predicted effects of our selective calpain-2 inhibitors on learning and memory and on neuroprotection have been demonstrated in several models in both rodents and NHP.

### 1. Acute glaucoma in mice.

We showed that NA-101 protected retinal ganglion cells (RGCs) and preserved vision following acute increase in intraocular pressure. These effects were observed with systemic administration of NA-101 as well as with intraocular administration. In both cases, the inhibitor prevented calpain activation as demonstrated by evaluation of the cleavage of known calpain targets, spectrin and PTEN. In contrast, RGC death was enhanced in calpain-1 knock-out mice.

### 2. Traumatic Brain Injury (TBI).

We showed that NA-101, NA-112 and NA-184 dose-dependently inhibited neuronal death in a mouse model of severe TBI, when administered systemically. In parallel, the inhibitors prevented calpain activation as demonstrated by evaluation of the cleavage of known calpain targets,

spectrin and PTEN. The inhibitors also prevented the increase in plasma levels of a fragment of PTPN13 generated by calpain-2 activation in the brain. NA-101 was also shown to be very effective in a mouse model of repeated mild concussion, mimicking the human disorder observed in many athletes, chronic traumatic encephalopathy (CTE). The extent of neuronal damage was also decreased in calpain-2 knock-out mice and exacerbated in calpain-1 knock-out mice, further demonstrating the requirement for a selective calpain-2 inhibitor. Similar effects were also found in a rat model of TBI. In all these models, the levels of calpain-2 activation were directly related to the extent of neuronal death. Moreover, the duration of brain calpain-2 inhibition by far exceeded the half-life of the inhibitor, an effect likely due to the formation of a slowly reversible covalent bond between the inhibitor and the active site of calpain-2.

### **3. Brain pathology following status epilepticus.**

We showed that calpain-2 was activated for a long period of time following status epilepticus, and this was associated with neuronal death, brain inflammation and cognitive impairment. Treatment with NA-112 for 7 days following status epilepticus prevented all features of brain pathology. Likewise, no brain pathology was observed in calpain-2 knock-out mice, clearly indicating the selectivity of NA-112 for calpain-2.

### **4. Enhanced cognition in mice and NHP.**

We showed that systemic administration of NA-101 could enhance learning and memory in normal mice at doses inhibiting calpain-2 while it reduced learning at doses inhibiting calpain-1. Calpain-1 knock-out mice showed learning impairment while calpain-2 knock-out mice showed enhanced learning and memory. Finally, acute systemic administration of NA-184 in aged macaques resulted in enhanced cognition. Chronic administration of NA-184 for 2 months resulted in continuous improvement of cognition without producing negative side-effects.

### **5. NA-184 stimulates neurogenesis in adult mice.**

We recently found that systemic injection of NA-184 in young adult mice stimulates neurogenesis in the subventricular zone and the dentate gyrus of the hippocampus by enhancing levels of a calpain-2 target, the transcription factor MEIS2. Further supporting these effects, we found that MEIS2 levels are enhanced in brains of calpain-2 knock-out mice, while they are decreased in brains of calpain-1 knock-out mice (manuscript in preparation).

To summarize, our selective calpain-2 inhibitors inhibit brain calpain-2 when administered systemically (ip, subcutaneously, iv, and possibly in). They provide cognitive enhancement and protect against neuronal death following various types of insults. This body of work makes a compelling argument for testing in humans to treat clinical indications associated with neuronal damage. Based on our assessment, an excellent clinical indication would be subarachnoid hemorrhage-induced vasospasm and brain complications.

## **Rationale for SAH-Induced Vasospasm as the First Clinical Indication**

We previously discussed testing NA-184 in 3-4 indications:

- SAH-induced vasospasm
- Neonatal Hypoxia/Ischemia Encephalopathy (HIE)
- Seizure-induced brain damage
- Acute glaucoma

We consulted with several clinicians specializing in these various fields. While we initially thought that acute glaucoma could provide a rapid proof-of-concept for the efficacy of our inhibitors, we realized that in fact, a clinical trial would take several years to demonstrate a beneficial effect, therefore ruling this indication out.

Seizure-induced brain damage remains an interesting potential indication, but there are not enough data indicating that brain imaging in the days following a seizure occurrence could provide a significant clinical outcome.

Neonatal HIE is also interesting, but there are many concerns with this indication, including variability of outcomes, combination with hypothermia, and difficulties translating safety data from adult to neonates.

We concluded that SAH-induced vasospasm is the most reasonable indication to obtain a proof-of-concept. As Michel indicated, he has a friend neurosurgeon in the Sacramento area, who could serve as a Principal Investigator for the clinical trial. He indicated that he can recruit at least 3 hospitals to participate in such a trial. He has prepared the attached details for the Phase 1 and Phase 2 for such a trial.

## **Rationale for Spinal Cord Injury (SCI)**

There is also a strong rationale to develop NA-184 for the treatment of SCI. The mechanism of axonal damage and neuronal injury is similar to that for TBI and there is strong evidence that calpain is activated following SCI. As discussed above, calpain-2 activation is directly linked to axonal and neuronal damage, and NA-184 is a potent and selective calpain-2 inhibitor. Clinical trials would be relatively easier and shorter than for TBI, as the clinical outcomes are more easily quantified than for TBI and occur shortly after the injury. We already talked to the Canadian Spinal Research Organization (CSRO), and they indicated that they would be interested in talking to us about a clinical trial for NA-184.

## **Additional (non-Neurodegeneration) Opportunities: Positive results with NA-184 in a Mouse Model of Cardiac Hypertrophy Following Heart Failure**

We worked with Dr. Javier Inserte at the Vall d'Hebron Institut de Recerca in Barcelona, Spain, who is studying the role of calpain in cardiac remodeling following heart failure. He obtained

compelling data indicating positive results with a 2-week treatment with NA-184 following heart failure to prevent cardiac hypertrophy. (Data attached).

We have filed a Provisional Patent to protect this novel application for our calpain-2 inhibitors.